Experiment-6 Parseval's Relation and Harmonic Spectrum Analysis

OBJECTIVE:

(i)To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.

(iv) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

(i) To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.

Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
clc;
clear;
close all;
figure;
f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0);
% Time-domain average power
P_time = (1/T)*trapz(t,abs(x).^2);

% Fourier-series coefficients
N = 20;
a0 = (2/T)*trapz(t,x);
```

```
an = zeros(1,N);
bn = zeros(1,N)
for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end

% Frequency-domain power
P_freq = a0^2/4 + 0.5*sum(an.^2 + bn.^2);

% Display results
fprintf('Time-domain power      = %.6f\n',P_time);
fprintf('Frequency-domain power = %.6f\n',P_freq);
fprintf('Absolute error         = %.6e\n',abs(P_time-P_freq));

% Comparison graph
bar([P_time P_freq]);
set(gca,'XTick',1:2);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power (W)');
title('Verification of Parseval''s Relation');
grid on;

clc;
clear;
close all;
```

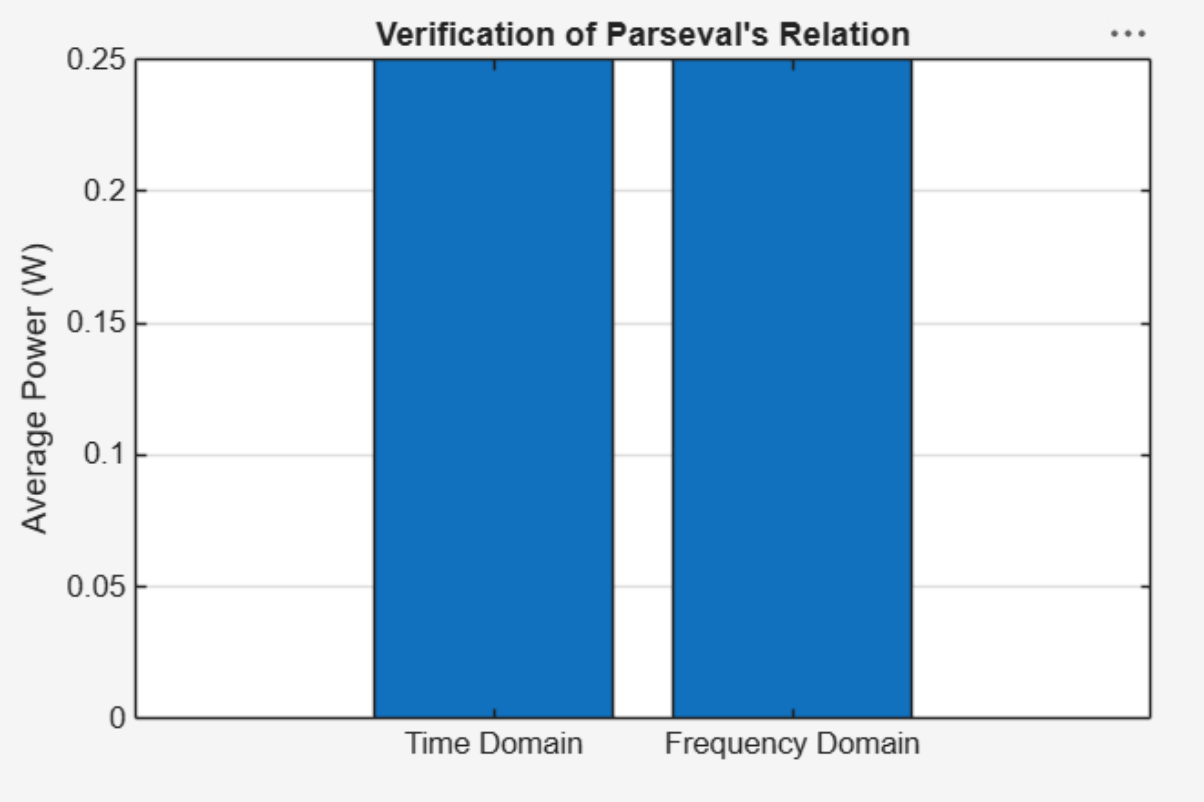
```
% Experiment 6(i): Verification of Parseval's Relation
figure;

f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0)
% Time-domain average power
P_time = (1/T)*trapz(t,abs(x).^2);
% Fourier-series coefficients
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a0 = (2/T)*trapz(t,x);
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% Frequency-domain power
P_freq = a0^2/4 + 0.5*sum(an.^2 + bn.^2)
% Display results
fprintf('Time-domain power      = %.6f\n',P_time);
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fprintf('Absolute error         = %.6e\n',abs(P_time-P_freq));
% Comparison graph
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bar([P_time P_freq]);
set(gca,'XTick',1:2);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power (W)');
title('Verification of Parseval''s Relation');
grid on;
```

OUTPUT:



(iv) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

Software / Laboratory Environment :

MATLAB

MATLAB Coding:

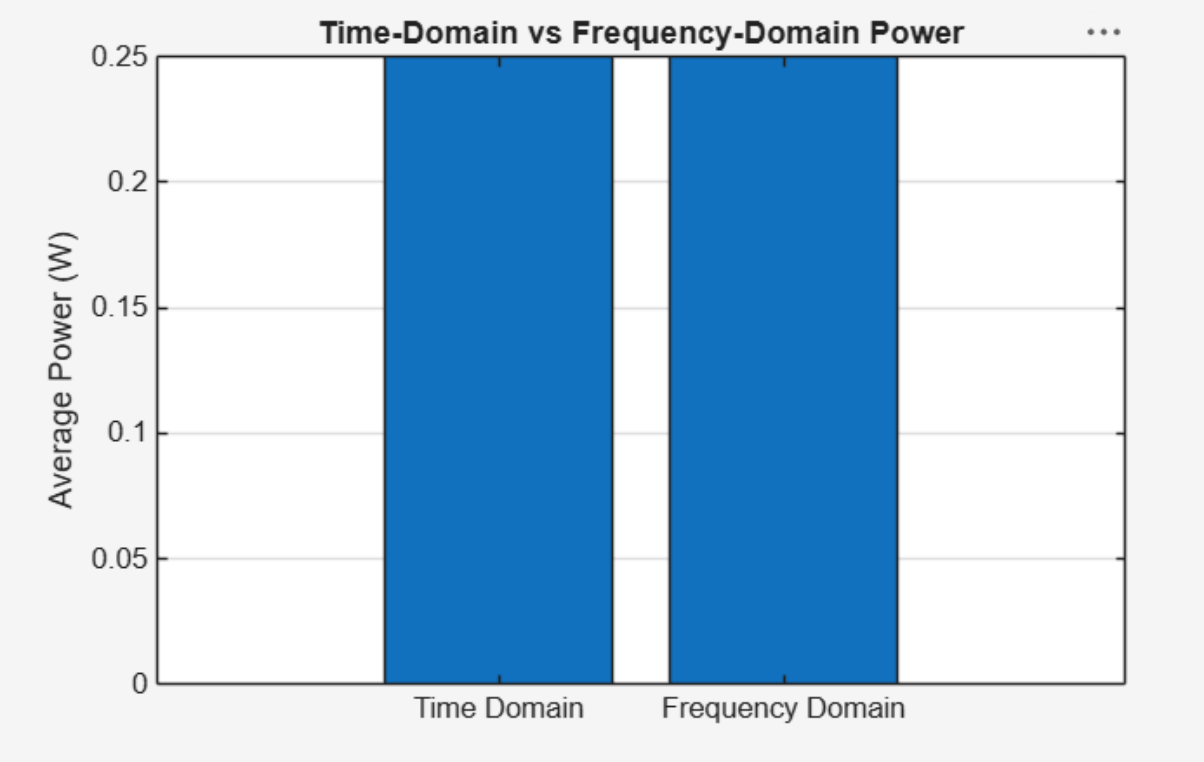
```
clc;
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```
clear;
close all;
figure;
f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0);
% Time-domain average power
P_time = (1/T)*trapz(t,x.^2);
% Fourier-series coefficients
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N);
bn = zeros(1,N)
for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end

% DC and harmonic power
P_dc = a0^2/4;
P_harmonics = 0.5*sum(an.^2 + bn.^2);
P_frequency = P_dc + P_harmonics
% Display results
fprintf('Time-domain power      = %.6f W\n',P_time);
fprintf('DC power               = %.6f W\n',P_dc);
```

```
fprintf('Harmonic power          = %.6f W\n',P_harmonics);
fprintf('Frequency-domain power = %.6f W\n',P_frequency);
fprintf('Absolute error          = %.6e W\n',abs(P_time-P_frequency))
% Comparison graph
bar([P_time P_frequency]);
set(gca,'XTick',1:2);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power (W)');
title('Time-Domain vs Frequency-Domain Power');
grid on;
```

Output Graphs:



RESULT:

The experiment successfully verified Parseval's relation for continuous-time periodic signals by demonstrating the equivalence of power calculated in the time and frequency domains. The distribution of power among the fundamental and harmonic components was analyzed through the harmonic spectrum, and the total

time-domain power was found to be consistent with the sum of harmonic powers in the frequency domain.

RUBRICS:

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**	**Marks(10)**	**Marks obtained**
(i) To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-	**Fourier Series Coefficient Computation**	Correctly identifies the fundamental period T0 and frequency ω0 of the periodic signal. Computes the Fourier series coefficients (c_k, or a_k and b_k) accurately, analytically or in MATLAB, and plots the signal	Computes the coefficients correctly with minor errors in the period, limits, scaling or labelling.	Computes only some coefficients correctly, or has noticeable errors in T0, integration or code, and needs guidance.	Unable to determine the period or compute the Fourier series coefficients.	**20**	2	

domain power.		with its harmonic reconstruction with correct labels and scaling.						
	Time-Domain Power & Parseval Verification	Correctly computes the average power P .Applies Parseval's relation correctly, verifies that the two values agree within a small numerical error, and explains the time-frequency relationship clearly.	Computes the power and applies Parseval's relation correctly, with minor numerical or interpretation errors.	Computes the time-domain power or applies Parseval's relation only partly. Verification is incomplete or has noticeable errors, and needs guidance.	Unable to compute the power or verify Parseval's relation.	**20**	2	

(ii) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.	**Harmonic Power Distribution**	Correctly computes the power in each harmonic and the DC power. Plots the power spectrum (stem plot) with labelled axes, shows the cumulative power, and identifies the dominant harmonics accurately.	Computes the harmonic powers and plots the spectrum correctly, with minor errors in indexing, scaling or labelling.	Computes some harmonic powers but has noticeable errors in the spectrum or the DC and harmonic terms, and needs assistance.	Unable to compute or plot the harmonic power distribution.	**20**	2	
	Time vs Frequency Power Comparison	Compares total time-domain power with the sum of harmonic powers for an increasing number of harmonics. Reports the percentage error and	Compares the total powers correctly, with minor errors in the error calculation or limited explanation of	Makes a basic comparison with incomplete error analysis, or gives limited explanation, and needs guidance.	Unable to compare the time-domain and frequency-domain power correctly.	**20**	2	

		the power captured, explains convergence and the effect of truncation (for example for square or triangular waves), and draws correct conclusions.	convergence.					
Interpret the results for both objectives and maintain a proper technical record.	**Engineering Analysis, Record & Interpretation**	Provides technically accurate interpretation of the results, relating power to harmonic content and signal waveform. Maintains a complete record with aim, theory, MATLAB code, plots, observation	Maintains a proper record and gives mostly accurate interpretation, with minor omissions in explanation or presentation.	Record and interpretation are basic, with partial explanation or missing elements.	Incomplete record, limited or incorrect interpretation of the results.	**20**	2	

		s, result and conclusion, with clear organizatio n.						
					Total	**10 0**	**10**	